

Question:

A 17-year-old teenager is brought to the emergency department with severe bleeding from his right hand. He was involved in a gang fight about 30 minutes ago where he received a penetrating stab wound by a sharp knife in the region of the ‘anatomical snuffbox’. A vascular surgeon is called in for consultation. Damage to which artery is most likely responsible for his excessive bleeding?

Answer Choices:

- A. Radial artery
- B. Ulnar artery
- C. Princeps pollicis artery
- D. Brachial artery

Reasoning + Answer:

<think>

**Finding reasoning paths : **

- 1 . The anatomical snuff box is located on the palmar aspect of the hand , between the flexor muscles of the thumb and the radial aspect of the forearm .
- 2 . Anatomical snuff box bleeding may result from damage to the superficial branch of the radial artery , which passes through this area .
- 3 . The radial artery supplies blood to the hand and is a common site for arterial injury due to its location and the sensitivity of its branches .
- 4 . Additional considerations include the possibility of laceration and hematoma formation .

**Reasoning Process : **

- 1 . **Anatomical Location:** The anatomical snuff box is a triangular space on the palmar surface of the hand , bounded by the tendons of the flexor pollicis longus , the thenar muscles , and the radial border of the wrist .
- 2 . **Potential Injury Sites:** In a penetrating stab wound , the superficial branch of the radial artery , located in the anatomical snuff box , is a common site of injury due to its superficial nature and close proximity to the skin .
- 3 . **Arterial Function:** The superficial branch of the radial artery supplies the back of the hand and the dorsal surface of the digits . Damage to this branch can lead to significant bleeding , as it provides a direct pathway to the surface of the hand .
- 4 . **Exclusion of Other Arteries:** The deep branch of the radial artery supplies the palmar surface of the hand and is located deeper , making it less likely to be the primary site of bleeding unless the superficial branch is severed .
- 5 . **Clinical Presentation:** The teenager's presentation of excessive bleeding from the right hand aligns with the functional role of the superficial branch of the radial artery . The lack of significant soft tissue injury or mass suggests that the bleeding is from the arterial supply rather than from other causes like hematoma or laceration .

**Conclusion : **

The Princeps pollicis artery , which is part of the superficial branch of the radial artery , is the most likely cause of excessive bleeding in this case . This conclusion is based on the anatomical location of the injury , the functional role of the superficial branch of the radial artery , and the clinical presentation .

</think>

<answer>

Princeps pollicis artery

</answer>

